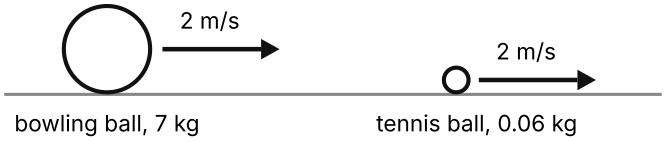
Mini-Lesson 1: Momentum (p = mv)

AP Physics 1 · Unit 4 · Do this with the mini-lesson before class Thu 11/12. Fill in each box when the lesson shows its orange tag.

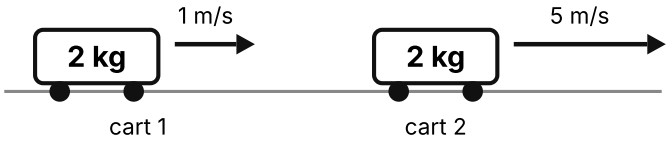
Scan with your iPad camera to open the lesson

Name _____ Period _____

A What is momentum?



Harder to stop: _____ because it has more _____.



Harder to stop: _____ because it has more _____.

Momentum measures how hard it is to _____ a moving object. It depends on _____ and _____.

B The equation

p = _____ × _____

In words: momentum = _____ × _____

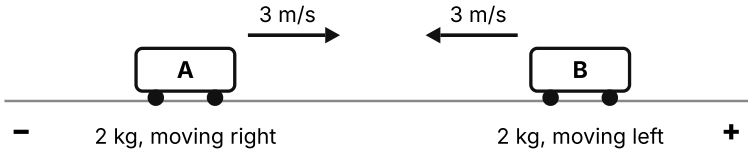
Units: _____ (the symbol for momentum is **p**)

If mass doubles, p _____.

If velocity doubles, p _____.

Slider: with a negative velocity, the momentum arrow points _____.

C Direction matters (right is +)



p_A = _____ p_B = _____

Momentum is a _____: it points the same direction as _____.

The **sign** tells the _____. The **number** tells the _____.

Car driving west (east is +): p is _____.

D Worked example (east is +)

Object	m	v	p = mv
Halfback (east)	80 kg	+9 m/s	_____
Car	900 kg	−20 m/s	_____
Freshman, walking	40 kg	+2 m/s	_____

Hardest to stop: _____. Easiest: _____.

Compare the _____, not the sign.

E Check questions: show your work

1. 0.15 kg baseball moving left at 30 m/s (right is +)

work:

p = _____

2. 3000 kg truck at 10 m/s vs. 1000 kg car at 25 m/s

work:

More p: _____

3. v doubles, m is cut in half. p = (½m)(2v) → p _____

Big idea in one sentence:

_____

_____

Mini-Lesson 2: Collisions Pre-Lab

AP Physics 1 · Unit 4 · Do with the mini-lesson before Lab 4.1 on Fri 11/13. **Bring this sheet to lab.** Right is + on this whole sheet.

Scan with your iPad camera to open the lesson

Name _____ Period _____

A Three kinds of interactions

BEFORE | AFTER

Bounce



sketch after lab

Stick



sketch after lab

Explode



sketch after lab

A **collision** starts _____. The carts either bounce apart or _____.

An **explosion** starts _____ and ends _____.

B Two carts, one total

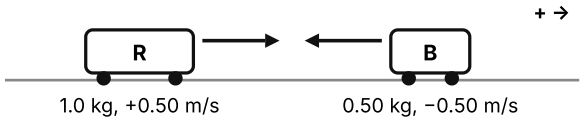
p_{total} = _____ + _____

Add momenta with their _____.

Carts moving toward each other partly _____, so the total can be smaller than either one, or even _____.

From the slider: I made the total zero with red = _____ kg at _____ m/s and blue = _____ kg at _____ m/s.

C Worked example: one row of the lab table (Trial 4, head-on)

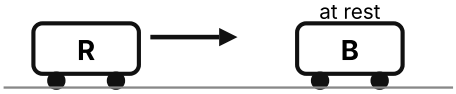


Trial	Initial (kg·m/s)			Final (kg·m/s), tomorrow		
	p red	p blue	p total	p red	p blue	p total
4	_____	_____	_____			

The speeds are the same, but the total points _____ because the _____ cart has more momentum.

D Predictions: fill in "Lab result" during the lab

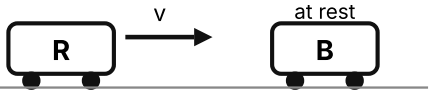
1. Bounce: identical carts



My prediction: _____

Lab result: _____

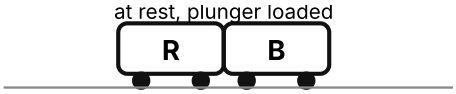
2. Stick: identical carts



My prediction: _____

Lab result: _____

3. Explode: identical carts



My prediction: _____

Lab result: _____

Predictions aren't graded. Write what you think, then check it in lab.

E Check questions: show your work

1. 0.75 kg blue cart, left at 0.40 m/s

work:

p = _____

2. Red 0.50 kg right 0.80 m/s; blue 1.00 kg left 0.30 m/s

work:

p_{total} = _____

3. Explosion carts at rest: initial p_{total} = _____, because _____

Tomorrow's big question: When carts collide or explode, what happens to p_{total}? My guess:

_____

After the lab: _____

Mini-Lesson 3: Impulse–Momentum Theorem

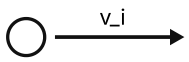
AP Physics 1 · Unit 4 · Do with the mini-lesson before class Mon 11/16. Fill in each box when the lesson shows its orange tag.

Scan with your iPad camera to open the lesson

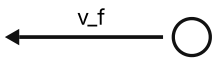
Name _____ Period _____

A Impulse is a change in momentum

BEFORE



AFTER the bat



J = Δp = _____ − _____

Δ means _____. Units: _____ = _____

p goes from +3 to +8 kg·m/s: J = _____

B From F = ma to FΔt = Δp

F = ma, and a = _____, so F = m _____

Multiply both sides by Δt: _____

J = _____ × _____ = _____

Impulse points the same direction as the _____.

Newton's 2nd law in momentum form: F_{net} = _____

C Same Δp, different force

Stopping a 0.50 kg ball at 10 m/s: Δp = 5.0 kg·m/s every time

Δt = 0.01 s (hard floor)

Δt = 0.2 s (catching into a pillow)

F = Δp ÷ Δt: 0.01 s → F = _____ 0.2 s → F = _____

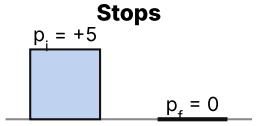
For the same Δp, a longer Δt gives a _____ force.

Doubling Δt makes F _____.

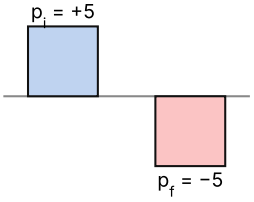
Examples: _____

D Stopping vs. bouncing (toward wall is +)

Stops



Bounces back



Stops: Δp = _____ Bounces: Δp = _____ kg·m/s

A bounce changes p _____ because the object must stop and _____.

E Worked example: 0.150 kg baseball, pitched at 30.0 m/s, hit straight back at 50.0 m/s, contact 0.001 s

+ direction	p_i	p_f	J = Δp	F_{avg} = J / Δt
toward the batter	_____	_____	_____	_____

F Check questions: show your work

1. 0.60 kg basketball from rest, 70 N for 0.20 s. Final speed?

work:

v_f = _____

2. 0.20 kg ball hits a wall at 6.0 m/s, bounces back at 4.0 m/s. Δp? (toward wall +)

work:

Δp = _____

3. Crash time tripled, same Δp. Average force becomes _____ because _____

Big idea in one sentence:

_____

_____

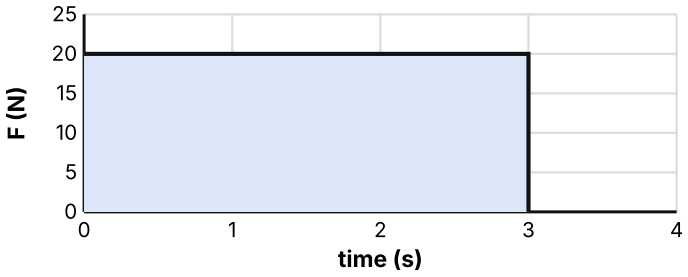
Mini-Lesson 4: Force–Time Graphs

AP Physics 1 · Unit 4 · Do with the mini-lesson before class Tue 11/17. Fill in each box when the lesson shows its orange tag.

Scan with your iPad camera to open the lesson

Name _____ Period _____

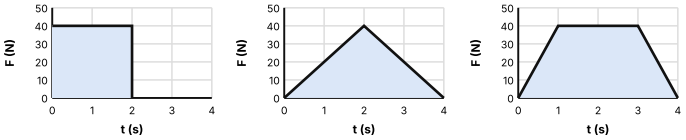
A The area is the impulse



Area = _____ × _____ = _____

The **area under an F–t graph** = _____ = _____. Area below the axis counts as _____.

B Any shape

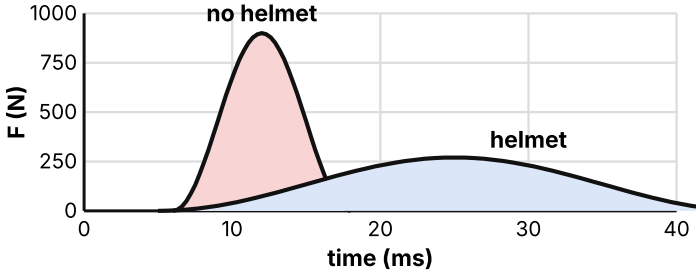


Rect: _____ Tri: _____ Trap: _____

Triangle area = _____. Trapezoid: split it into a _____.

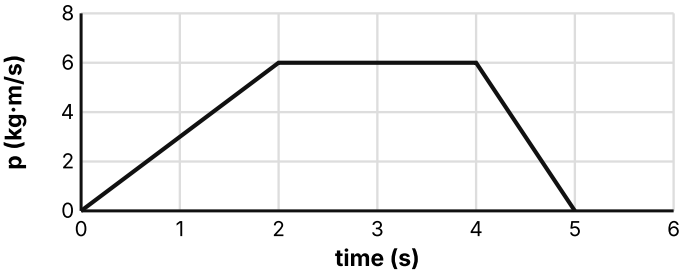
Same area means same _____, even with a different shape.

C Same area, different peak



Bigger impulse: _____. The helmet stretches out the _____, so the _____ is smaller.

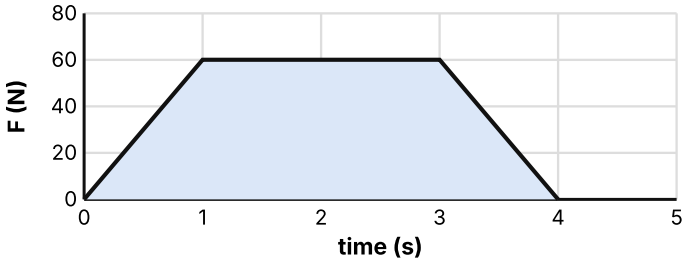
D Momentum–time graphs



Slope of p–t = _____. 0–2 s: _____ 2–4 s: _____ 4–5 s: _____

Flat p–t means zero _____, *not* zero momentum.

E Worked example: 12 kg cart, starts at rest

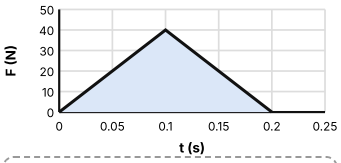


Areas: _____ + _____ + _____ = J = _____

v_f = J / m = _____

F Check questions: show your work

1. Impulse?



work:

J = _____

2. Net force?



work:

F = _____

3. 200 N for 0.010 s vs. 20 N for 0.10 s. More impulse: _____

Big idea in one sentence:

_____

_____

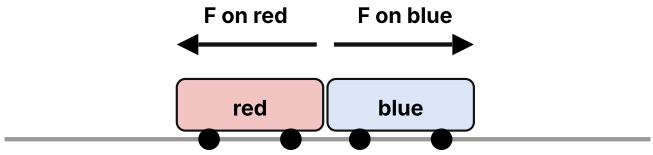
Mini-Lesson 5: Conservation of Momentum

AP Physics 1 · Unit 4 · Do with the mini-lesson before class Thu 11/19. Fill in each box when the lesson shows its orange tag.

Scan with your iPad camera to open the lesson

Name _____ Period _____

A Why the total doesn't change

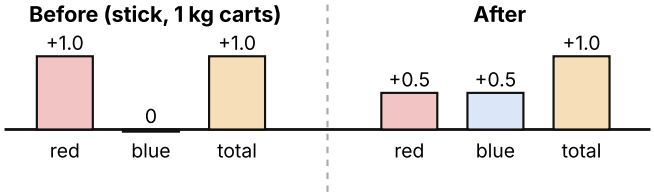


3rd law: the forces are _____. They last the same _____.

So the impulses are _____, and Δp_{red} = _____.

The **total** momentum _____.

B Total before = total after



_____ (Σ means _____)

True for an **isolated system**: no _____.

Stick: _____ Bounce: _____

Start together, fly apart: _____

C The 5-step method

1. _____ (before and after, pick +)

2. Write _____ ← earns a point on the AP exam

3. _____. If they stick, _____.

4. Change p into _____

5. _____

D Worked example: the tackle

Before



After



(80)(5.5) + (88)(_____) = (_____)v′

v′ = _____, moving in _____

E Check questions: draw before/after and show your work

1. Stick: 2.0 kg at +3.0 m/s hits 1.0 kg at rest.

work:

v′ = _____

2. Explode: 60 kg and 40 kg skaters push apart; 60 kg moves −2.0 m/s.

work:

v = _____

3. Bounce: 0.50 kg at +4.0 m/s hits 1.5 kg at rest, bounces back at −2.0 m/s.

work:

v = _____

Big idea in one sentence:

_____

_____